connected across its contacts. The npn-pnp junction (the Sziklai pair) offers an advantage over other Darlington pairs due to its lower base-emitter voltage of 0.61V, compared to 1.22V for other Darlington pairs, making low-voltage operation possible.

Calibration

This is where we eliminate the need for bulky MCUs and simplify the circuit. The 4.7k pot (VR1) is a precise multi-turn type used to set the temperature setpoint. Heat the probe to the desired temperature and then adjust pot VR1 so that the relay just de-energises. Beyond this point, if the temperature increases, the relay will energise again. The normally closed (NC) or normally open (NO) contact of the relay can be used to disconnect/reconnect a charger, gadget, cooker, soldering iron, etc. Conversely, when the temperature decreases, the TMP36 cools down, de-energising the relay and reconnecting the device.

Since the TMP36 has a ramp up of 3°C/second and a ramp down of 6°C/second, it automatically accounts for a 3-degree overlap per second to avoid oscillation over a mean temperature. Additionally, physical bodies do not dissipate heat quickly, unless fast cooling is provided.

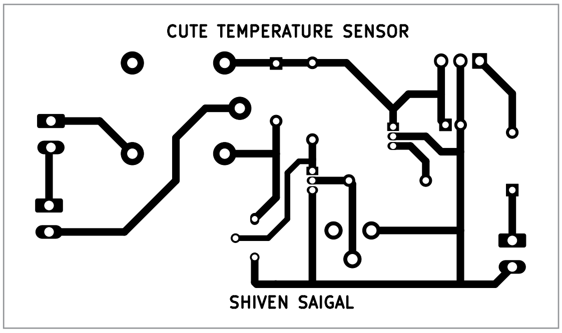

Fig. 4: Actual-size PCB layout of the circuit

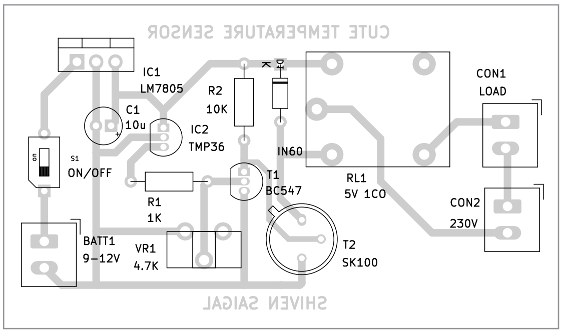

Fig. 5: Component layout of the PCB

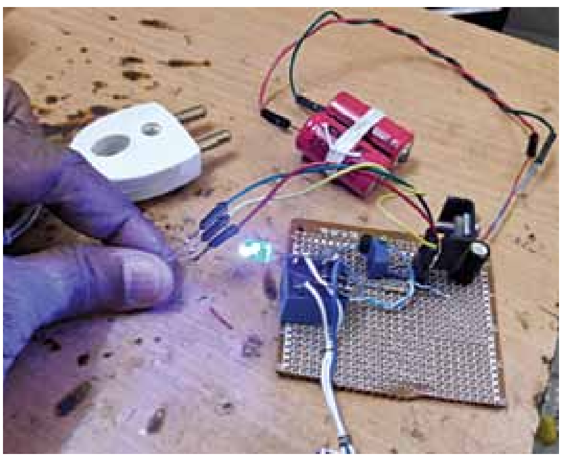

Fig. 6: Author's prototype of the cute tempearture sensor

Construction and testing

The actual-size, single-side PCB layout for the cute temperature sensor is shown in Fig. 4, while its component layout is shown in Fig. 5. After assembling the circuit on PCB, enclose it in a suitable box.

Alternatively, you can assemble the circuit on a general-purpose PCB, as shown in the author's prototype depicted in Fig. 6. Place all the components and the PCB in a suitable enclosure, and provide terminal strips for the 230V supply input and output separately.

PARTS LIST	
Semiconductors:	
IC1	- 7805, 5V regulator
IC2	- TMP36 temparature sensor
D1	- 1N60 diode
T1	- BC547 npn transistor
T2	- SK100 pnp transistor
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
R2	- 10-kilo-ohm
VR1	- 4.7-kilo-ohm pot
Capacitor:	
C1	- 10µF, 16V electrolyte
Miscellaneous:	
CON1, CON2	- 2-pin connector
RL1	- 5V, 1C/O relay
S1	- On/off switch
BATT.1	- 9-12V battery

This temperature sensor has various applications:

Charging Ni-Mh or Li-ion batteries. It can be used to charge Ni-Mh or lithium-ion (Li-ion) batteries. By preventing overcharging when the battery reaches a specific temperature (45°C to 50°C for Ni-Mh batteries and 45°C for Li-ion and Lipo batteries), the project extends the batteries' lifespan beyond normal usage.

Soldering iron tip temperature control. Use the NC connection to disconnect the soldering iron supply and position the TMP36 slightly away from the tip, where the temperature remains below 150°C.

Modulating the gas burner when water starts boiling. Use the normally open (NO) connection to regulate the shimmer knob/motor and place the TMP36 in direct contact with the boiling liquid, using a leak-proof metallic shield for the TMP36.

In a world driven by advanced technology, this temperature sensor reminds us that simplicity can be powerful. Streamlining temperature control with precision and ease opens doors to endless possibilities for enhancing our daily lives. EFY

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